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High-pressure structural phase transition and electronic properties of the alkali hydrides compounds XH (X = Li, Na)

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ABSTRACT

First principle calculations based on the density functional theory using the full-potential linearized augmented plane wave method have been carried out to determine the structural stability of different crystallographic phases, the pressure-induced phase transition and the electronic properties of LiH and NaH compounds. The rocksalt, zincblende, cesium chloride (CsCl) and wurtzite structures are considered. The exchange and correlation potential is treated by using the improved generalized-gradient approximation Moreover, the modified Becke–Johnson (mBJ) scheme is also applied to optimize the corresponding potential for the band structure calculations. The calculated ground state parameters for these compounds in each structure are well compared with the available theoretical and experimental results. The calculated band structures using the mBJ-GGA approach have an insulating nature for both compounds in all the considered structures, except the LiH for CsCl structure which shows a semi-conducting behavior. The results are in good agreement with other calculations and experimental measurements.

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FP-LAPW; phase transitions; LiH; NaH; mBJ potential

1. Introduction

The alkali hydrides XH (X = Li, Na, K, Rb and Cs) have been the subject of much attention due to their possible applications in nuclear and chemical industries [1]. These molecules attract researchers in various areas because they have the simplest electronic structure which allowing comparisons between different theoretical models. These materials have been investigated both experimentally and theoretically for many years. The X-ray experimental study shows that XH compounds crystallize in the rocksalt (RS) structure at normal conditions [2], while a structural phase transition from the RS (B1) to the CsCl (B2) phase was observed for CsH, NaH, KH and RbH compounds. The transition pressure for these compounds ranges from 1.2 GPa for CsH to approximately 30 GPa for NaH [3–5], while the B1 to B2 phase transition in LiH has not yet been observed. Hochheimer et al. [4] performed a high-pressure energy-dispersive X-ray study of the alkali hydrides NaH, KH, RbH, and CsH. The structural phase transition from B1 to B2 phase was observed for KH, RbH and CsH at high pressure, while this structural phase transition was not observed for NaH for a pressure up to 28.0 GPa. The authors [4] found that the transition pressure from the B1 to B2 phase structure

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decreases when increasing the alkali radii. This result has been suggested by Bashkin et al. [6], which depends on the assumption that the alkali hydrides and alkali halides show a similar behavior.

There have been many theoretical predictions for LiH phase transition and the estimated transition pressure ranging from about 200 to 600 GPa. Ahuja et al. used the full-potential linear-muffin-tin-orbital method to theoretically investigate the structure of both LiH and NaH under such high pressure [7]. Sudha et al. [8] investigated the structural, electronic and elastic properties of alkali hydrides XH (X = Li, Na, K, Rb, Cs) by using the Vienna *ab initio* simulation package. The calculated transition pressure from the B1 to B2 phase is predicted to be 208 GPa and for LiH and 37 GPa for NaH using PBE-GGA approach. Guangwei et al. [9] predicted that the structure of LiH will change from B1 to B2 at very high pressure, approximately at 660 GPa.

From the above it is clear that there is considerable works on the considered compounds, involving both experimental and theoretical methods, but there are no reported studies on the ground state parameters, electronic structure and structural phase transition for both the zincblende (ZB) and wurtzite (WZ) phases. The reasons mentioned above motivated us to perform such calculations on the structural, electronic and the structural phase transition for LiH and NaH in RS, CsCl, ZB and WZ phases, using the full-potential augmented plane wave method (FP-LAPW). This will provide reference data for the experimentalist and will serve as additional data to the existing theoretical works on these compounds.

2. Method of calculation

The calculations reported here on the XH compounds in the (RS), (CsCl), (ZB) and (WZ) structures were performed using the FP-LAPW method within the frame work of the density functional theory (DFT) as implemented in the WIEN2K code. This is considered to be one of the most efficient methods of simulating and calculating the ground state properties of crystalline materials [10–12]. In this method, the unit cell is divided into two parts to perform the calculations: non-overlapping atomic spheres and interstitial region. The generalized gradient approximation (GGA) [13] has been used to calculate the structural properties, while the modified Becke–Johnson (mBJ) formalism is used to overcome the well-known underestimation of the band gap values by the GGA method [14].

Calculations based on the DFT within the local density approximation (LDA) and PBE-GGA approximations are accurate and helpful for the interpretation of experimental data. However, the major drawback of the DFT formalism with the above approximations is the improper interpretation of the excited state properties shown by the severe underestimation of the band gap value and the overestimation of the electron delocalization, particularly for systems with localized *d* and *f* electrons. Actually some approximations beyond the LDA and PBE-GGA, such as GW, hybrid functional, LDA + U etc., are developed in order to describe accurately the electronic structure of semiconductors and insulators [15,16]. However, in all cases, these methods can be computationally expensive or unsatisfactory. For example, the LDA + U method can only be applied to correlated and localized electrons. Fortunately, the recently proposed mBJ potential approximation is an alternative way to have a band gap close to the experimental value, as well as being computationally cheaper than the other aforementioned methods.

The electronic structure can be performed with the Kohn–Sham (KS) equations given by this relation:

$$\left(-\frac{\nabla^2}{2}+V_{\mathrm{eff},\sigma}^{\mathrm{KS}}(r)\right)\psi_{i,\sigma}(r)=\varepsilon_{i,\sigma}\ \psi_{i,\sigma}(r) \tag{1}$$

where $\psi_{i,\sigma}$ is the one electron wave functions, $\varepsilon_{i,\sigma}$ is the electron energies and $V_{\mathrm{eff},\sigma}^{\mathrm{KS}}$ is the KS effective potential, including the exchange correlation potential approximated by PBE-GGA and LDA

approximations. The mBJ potential is expressed as follows:

$$V_{x,\sigma}^{\mathrm{mBJ}}(r) = cV_{x,\sigma}^{\mathrm{BR}}(r) + (3c-2)\frac{1}{\pi}\sqrt{\frac{5}{6}\left(t_\sigma(r)/\rho_\sigma(r)\right)} \tag{2}$$

where the corresponding electron density and kinetic energy-density are given as

$$\rho_\sigma = \sum_{i=1}^{N_\sigma} |\psi_{i,\sigma}|^2 \tag{3}$$

$$t_\sigma = \left(\frac{1}{2}\right)\sum_{i=1}^{N_\sigma} \vec{\psi}_{i,\sigma}^{*} \cdot \vec{\psi}_{i,\sigma} \tag{4}$$

and $V_{x,\sigma}^{\mathrm{BR}}$ is the Becke–Roussel exchange potential [17], given by the following relation:

$$V_{x,\sigma}^{\mathrm{BR}}(r) = \frac{-1}{b_\sigma(r)}\left(1 - e^{-x_\sigma(r)} - \frac{1}{2}x_\sigma(r)e^{-x_\sigma(r)}\right) \tag{5}$$

x_σ can be found from a nonlinear equation involving t_σ, ρ_σ, $\vec{\nabla}\rho_\sigma$ and $\nabla^2\rho_\sigma$.

The $b_\sigma(r)$ is equal to $\left[x_\sigma^3 e^{-x_\sigma}/(8\pi\rho_\sigma)\right]^{1/3}$, c is chosen to depend linearly on the square root of the average of $\frac{|\rho|}{\rho}$

$$c = A + B\sqrt{\frac{1}{V_{\mathrm{cell}}}\int \frac{|\vec{\nabla}\rho_\sigma(r')|}{\rho_\sigma(r')}\,\mathrm{d}r'} \tag{6}$$

where A and B are two parameters with values $A = -0.012$ and $B = 1.023$ Bohr$^{1/2}$, according to the best fit from the experimental results of the semiconductor band gaps, and V_{cell} is the unit cell volume.

Spherical harmonics expansion is used inside the muffin tin (MT) spheres, and the plane wave basis set is used in the interstitial part. The muffin tin radii (R_{MT}) used in the present calculations for the H, Li and Na atoms are 1.1, 2.0, and 2.56 atomic units (a.u.), respectively. The charge density was Fourier expanded up to $G_{\max} = 20$ in RS, CsCl and ZB, while $G_{\max} = 14$ in WZ structure. The plane wave cut-off was taken such as that the cut-off $R_{\mathrm{MT}}.K_{\max} = 5$.

For energy convergence, the full Brillouin zones were sampled with 1331 k-points for the three structures, RS, ZB and CsCl. In the irreducible Brillouin zone (IBZ), a grid size of 11 × 11 × 11 was used, which is reduced to 56 special k-points. For the WZ structure, a grid size of 18 × 18 × 18 was used with 3700 k-points, which is then reduced to 222 special k-points in the IBZ [18]. The self-consistent calculation of the total energy of the unit cell was converged to less than 10^{-5} Ry/unit cell.

3. Results and discussion

3.1. Structural properties

The ground state properties for each structure are obtained by calculating the total energy per unit cell at several volumes around the equilibrium volume and fitting the calculated values into the

Birch–Murnaghan equation of state [19] using the following relation:

$$E\ (V) = E_0\ + \frac{9V_0B_0}{16}\left\{\left[\left(\frac{V_0}{V}\right)^{\frac{2}{3}} - 1\right]^3 B_0^{'} + \left[\left(\frac{V_0}{V}\right)^{\frac{2}{3}} - 1\right]^2\left[6 - 4\left(\frac{V_0}{V}\right)^{\frac{2}{3}}\right]\right\} \tag{7}$$

where E_0 is the minimum energy, V_0 is the equilibrium volume, B_0 is the bulk modulus and

$$B_0^{'} = dB_0/dP \tag{8}$$

The accurate pressure corresponding to the desired volume is calculated using the following relation:

$$P\ (V) = \frac{3B_0}{2}\left[\left(\frac{V_0}{V}\right)^{\frac{7}{3}} - \left(\frac{V_0}{V}\right)^{\frac{5}{3}}\right]\left\{1 + \frac{3}{4}(B_0^{'} - 4)\left[\left(\frac{V_0}{V}\right)^{\frac{2}{3}} - 1\right]\right\} \tag{9}$$

The calculated structural parameters such as the lattice constant a_0, bulk modulus B_0, and the pressure derivatives of the bulk modulus $B_0^{'}$ for LiH and NaH in RS, CsCl and ZB phases are listed in Tables 1–3, respectively, along with other experimental and theoretical works. The calculated structural parameters for LiH and NaH in RS structure are shown in Table 1, which are in agreement with the previous experimental results [3,4,5,20–22] and theoretical work [8,23–28]. There is an agreement between the predicted values of B_0 and $B_0^{'}$ for NaH and the experimental results [5]. Moreover, Table 2 shows that the calculated structural parameters for LiH and NaH in CsCl structure are in agreement with other theoretical results [8,27,29–32]. The calculated structural parameters for LiH and NaH in the WZ structure are presented in Table 4. One can see that the c/a_0 ratio increases as the alkali atomic number increases; it is roughly 1.49 for LiH and 1.55 for NaH.

Table 1. Structural parameters for LiH and NaH in RS structure along with experimental and other theoretical results.

Compound	Structural parameters	Present calculations	Experimental results	Other theoretical results
LiH	a_0 (Å)	4.018	4.075[20], 4.084[21]	4.0811[8], 3.92[23], 4.03[24]
	B_0 (GPa)	36.85	32.2[3]	33[8], 31[24], 34.1 [25], 32.3[26]
	$B_0^{'}$	4.02	3.95[22]	4.9[8], 3.5[24]
NaH	a_0 (Å)	4.838	4.880[20]	4.8511[8], 4.775[23], 4.921[24] 4.865[27]
	B_0 (GPa)	23.30	19.4±2[5],14.3 ± 1.5[4]	26[8], 29.6[23], 20[24], 27.4[28] 22.90[27]
	$B_0^{'}$	3.55	4.40 ± 0.5[5], 7.7 ± 1.0[4]	3.62[8], 4.1[24], 3.78[27]

Table 2. Structural parameters for LiH and NaH in CsCl structure along with experimental and other theoretical results.

Compound	Structural parameters	Present calculations	Experimental results	Other theoretical results
LiH	a_0 (Å)	2.510	————	2.458[8]
	B_0 (GPa)	33.60	————	30.16[8]
	$B_0^{'}$	4.01	————	4.05[8]
NaH	a_0 (Å)	2.965	3.094[5]	3.010[8], 4.838[29], 4.955[30]
	B_0 (GPa)			2.982[27]
	$B_0^{'}$	23.97	28.30 ± 3.0[5]	28[8], 22.81[29] 23.5[31], 23.21[27]
		3.83	4.30 ± 0.40[5]	2.669[8], 3.75[32], 3.16[31], 3.75[27]

Table 3. Structural parameters for LiH and NaH in ZB structure.

Compound	a_0 (Å)	B_0 (GPa)	$B_0^{'}$
LiH	4.307	27.17	3.94
NaH	5.228	17.49	3.67

Table 4. Structural parameters for LiH and NaH in WZ structure.

Compound	a_0 (Å)	c/a_0	u (a.u.)	B_0 (GPa)	B_0'
LiH	3.115	1.4942	0.385	27.91	3.36
NaH	3.748	1.5672	0.390	18.06	3.90

3.2. Phase transition

The energy per unit cell as a function of volume is calculated by using both the PBE-GGA and mBJ-GGA methods. The total of energy versus cell volume, calculated using PBE-GGA for LiH and NaH, is shown in Figures 1 and 2, respectively. It is apparent from these two figures that the RS structure, at ambient pressure, is thermodynamically stable structure (ground state structure) for both LiH

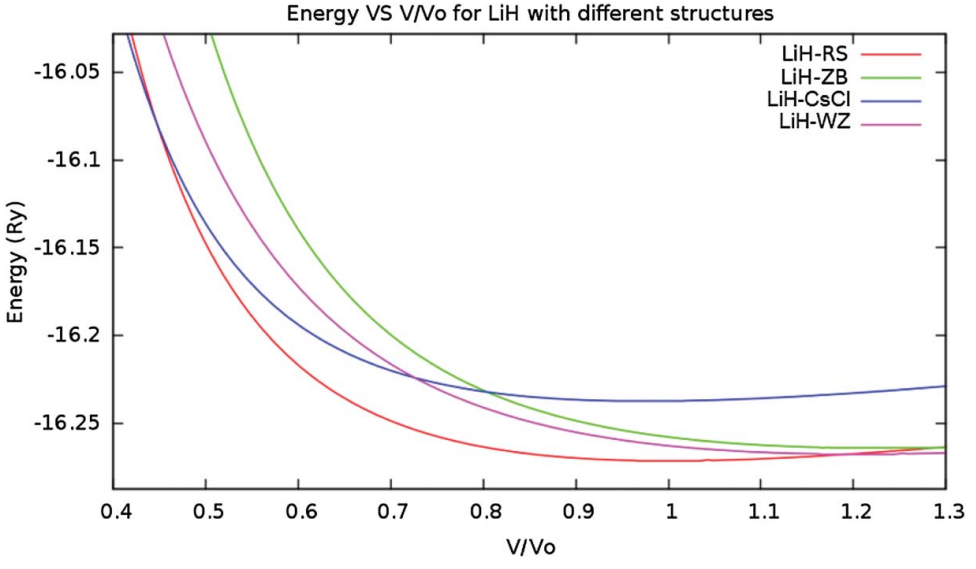


Figure 1. (Color online) Calculated total energy per unit cell versus V/V_0 for LiH with RS, CsCl, ZB and WZ structures.

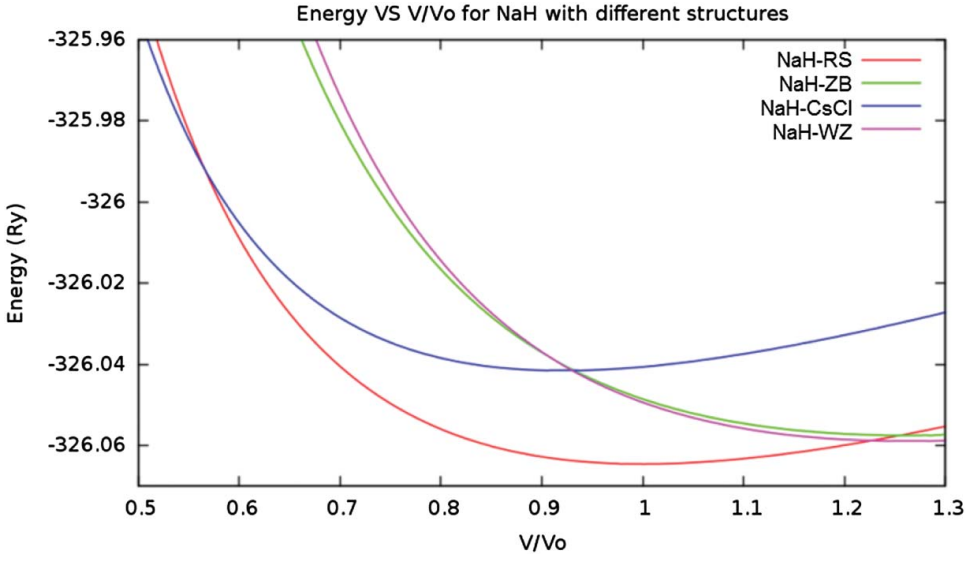


Figure 2. (Color online) Calculated total energy per unit cell versus V/V_0 for NaH with RS, CsCl, ZB and WZ structures.

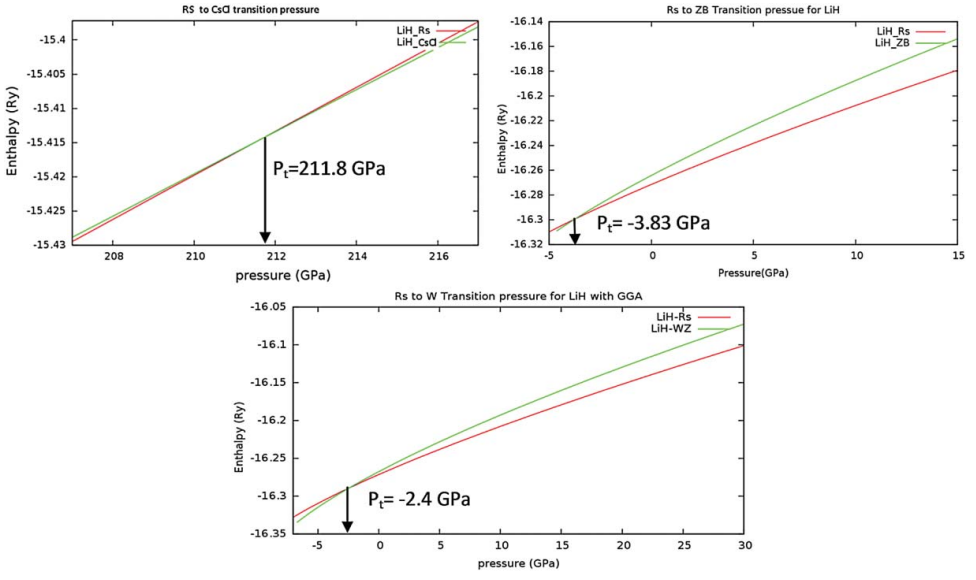


Figure 3. Enthalpy as a function of pressure for LiH using PBE-GGA approximation.

and NaH. Under compression, the calculations show that both compounds undergo a structural phase transition from RS to CsCl structures, while under expansion the structures undergo a transition from RS to ZB or from RS to WZ. The structural phase transition is determined by calculating the Gibbs free energy (G) for the two phases, which is given by $G = E_0 + \mathrm{PV} - \mathrm{TS}$. Since the theoretical calculations are performed at $T = 0$ K, Gibbs free energy becomes equal to the enthalpy, $H = E_0 + \mathrm{PV}$. For a given pressure, a structure is a stable structure when the enthalpy is at its *lowest* value. The enthalpy versus pressure curves for the both structures for LiH and NaH are displayed in Figures 3 and 4. The transition pressure P_t can be estimated by using the usual condition of equal enthalpies for the two phases at the point of intersection. From the common tangent technique, we can also calculate the phase transition pressure which is equal to the negative of the slope tangent line to the curves in Figures 1 and 2.

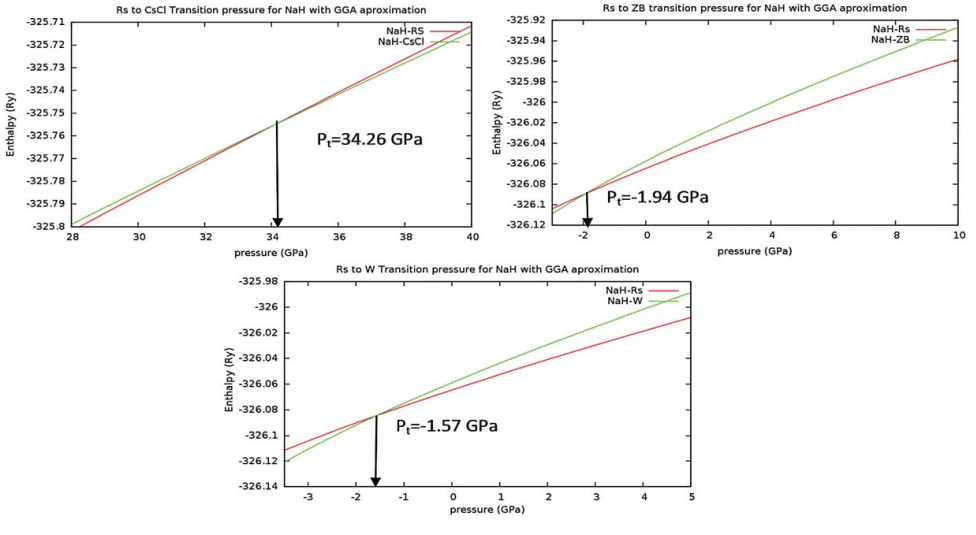


Figure 4. Enthalpy as a function of pressure for NaH using PBE-GGA approximation.

Table 5. The V/V_0 fraction for LiH and NaH.

Compound	RS → CsCl	RS → ZB	RS → WZ
V/V_0 for LiH	0.446	1.291	1.194
V/V_0 for NaH	0.565	1.255	1.225

Table 5 shows the V/V_0 fractions, where V_0 is the equilibrium volume of the RS unit cell and V is the transition volume. The calculated transition pressure from RS to CsCl structure for NaH is 34.26 GPa, which is in line with the diamond anvil-cell high-pressure experimental value of about 30 GPa [5], and in the range of the previous theoretical calculations [7,8,9,27,29]; whereas, for LiH compound the calculated transition pressure is 211.8 GPa using PBE-GGA approach, which is in the range of the previous theoretical calculations [8,33]. The transition from RS to CsCl for both LiH and NaH occurred at very high pressure with $V/V_0 = 0.446$ and 0.565 for LiH and NaH, respectively. It is due to a volume compression of the RS lattice with respect to the equilibrium volume, which corresponds to a decrease of the optimized lattice constant of 23.6% and 17.3% for LiH and NaH compounds, respectively.

The transition pressure from RS to WZ or ZB phases requires volume expansion because we got a negative pressure, which means we have to expand the RS structure to get the WZ or ZB phase. The calculated transition pressure from RS to WZ phase for both LiH and NaH are −2.4 and −1.57 GPa, respectively and occurred at low pressures with $V/V_0 = 1.194$ and 1.225 for both LiH and NaH, respectively. This is because one has to expand the RS lattice parameter to WZ by about 6.1% for LiH and 7% for NaH, while the calculated transition pressures from RS to ZB structure for both LiH and NaH are −3.83 and −1.94 GPa, respectively and occurred at low pressures with $V/V_0 = 1.291$ and 1.255 for both LiH and NaH, respectively. This is because one has to expand the RS lattice parameter to ZB by about 8.9% for LiH and 7.9% for NaH. It is now obvious that both LiH and NaH undergo a structural phase transition at a high pressure from RS structure to CsCl structure. On contrast, under lower pressure the RS structure expands to larger volume and transforms to WZ phase or ZB phase. There is about a 4 GPa (13%) difference in the computed transition pressure for the NaH compound and the experimental value [5]. This is mainly related to two main reasons: the purity of powder samples used in the experiments and the difference in the temperatures used in the experimental and theoretical calculations. The experiments were carried out at room temperature, while theoretical calculations are carried out at $T = 0$. The computed transition pressures, as well as the previous theoretical [7,8,9,27,29,33] and experimental data for LiH and NaH, are given in Table 6. From these two tables, it is evident that the RS to CsCl transition pressure increases as the alkali radii decreases, as has been proposed by Bashkin et al. [6]. On the other hand, the RS to ZB and the RS to WZ negatively increases as the alkali radii decreases.

3.3. Electronic band structure

The calculated band structures along the high symmetry lines in the Brillouin zone, for LiH and NaH in the RS, CsCl, ZB and WZ phases using the PBE-GGA and mBJ-GGA approaches for the exchange–correlation potential, are shown in Figures 5 and 6. The band structures are calculated using the computed equilibrium lattice constants. Figures 5 and 6 show that the LiH in the RS structure has a direct energy band gap with the valence and conduction bands that are situated at the X-point symmetry line, while the other phases have an indirect energy band gap.

Table 6. Calculated transition pressure P_t as well the experimental and other theoretical data for LiH and NaH compounds.

	RS → CsCl P_t (GPa)			RS → ZB	RS → WZ
Compound	Present work	Experimental works	Other theoretical works	P_t (GPa)	P_t (GPa)
LiH	211.8	————	208 [8], 329[33], 660[9]	−3.83	−2.4
NaH	34.26	29.3±0.9 [5]	37 [7, 8, 9], 32[27, 29]	−1.94	−1.57

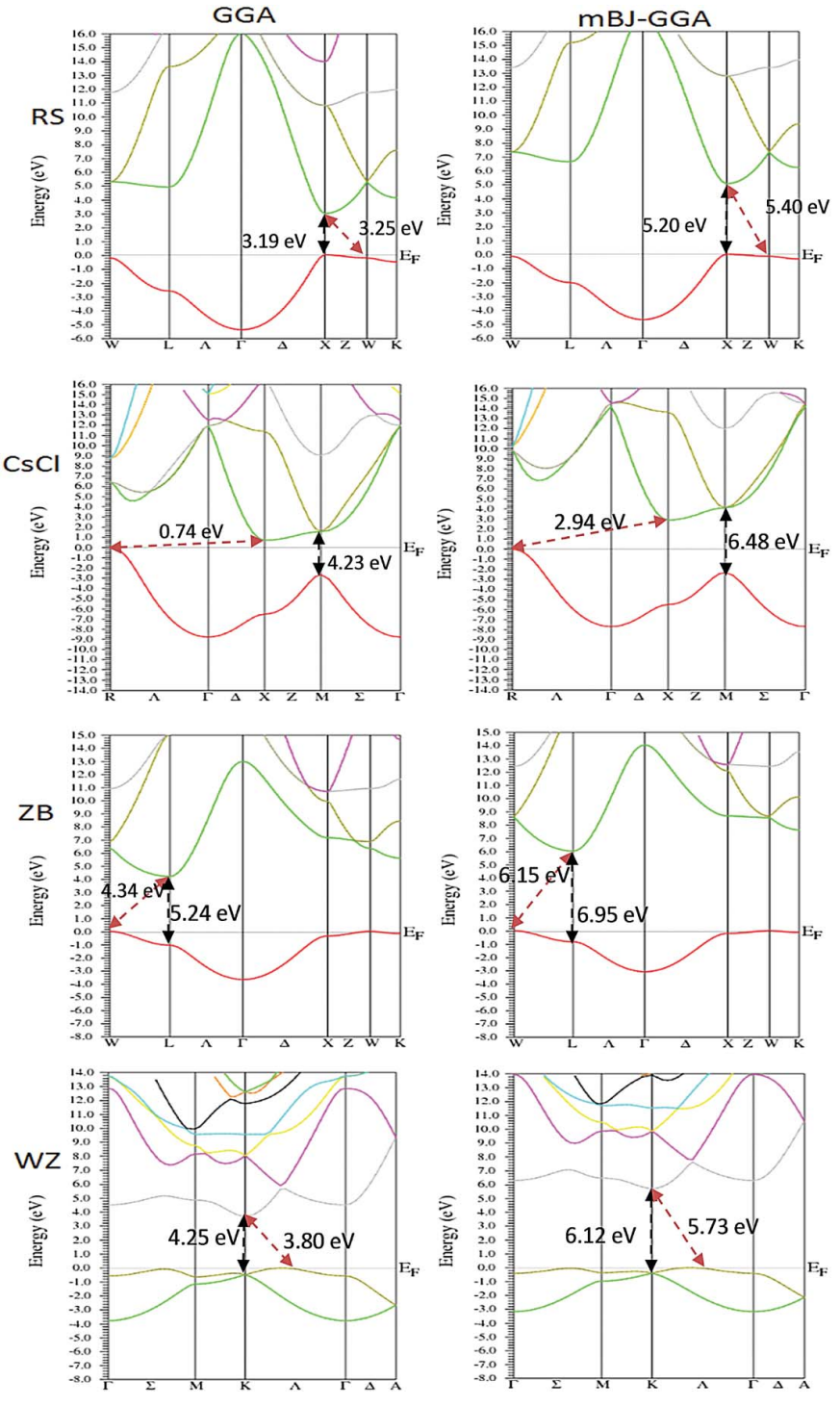


Figure 5. Band structure of LiH in RS, CsCl, ZB and WZ with PBE-GGA and mBJ-GGA approaches.

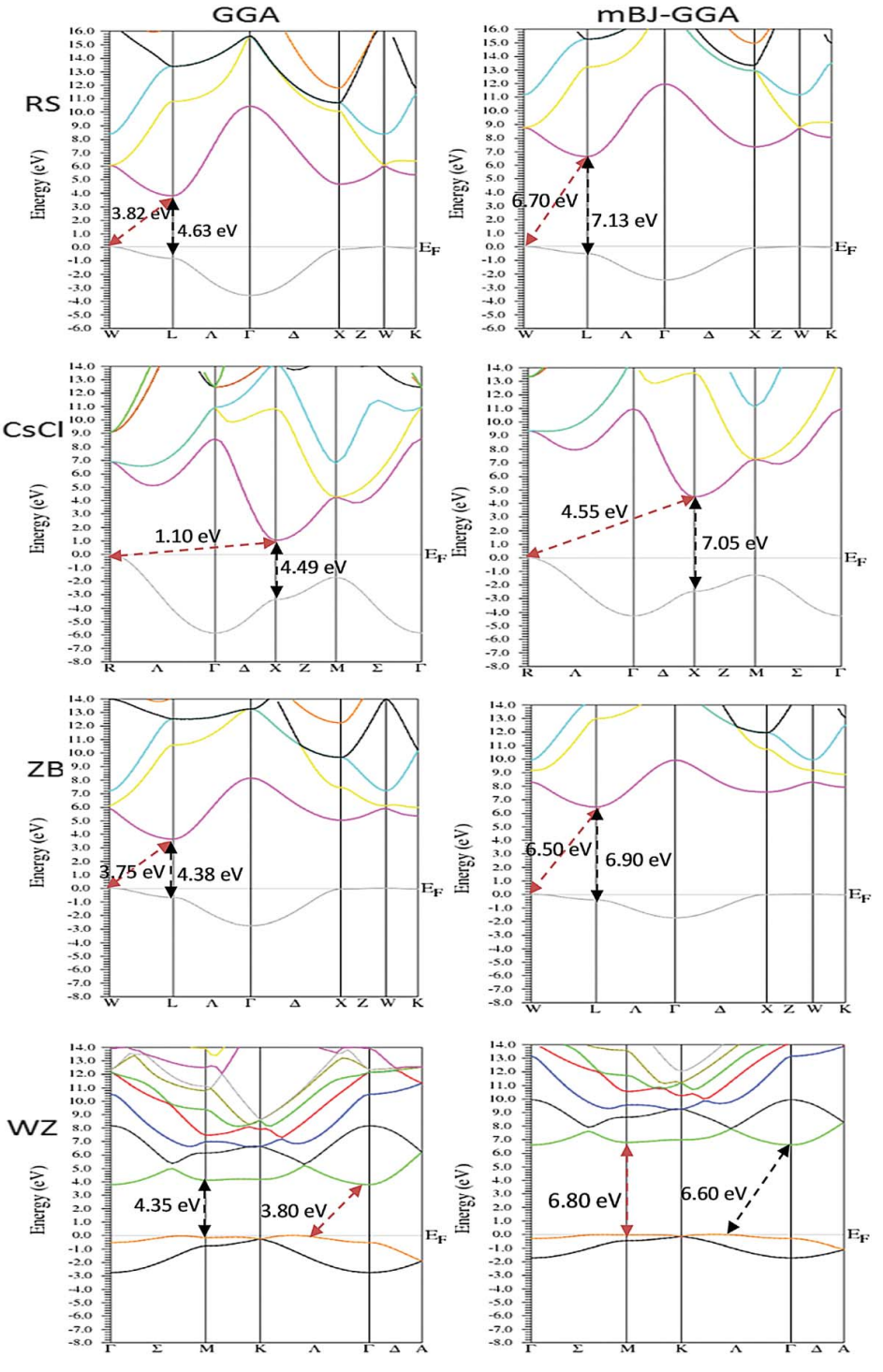


Figure 6. Band structure of NaH in RS, CsCl, ZB and WZ with PBE-GGA and mBJ-GGA approaches.

Table 7. Calculated energy band gap value E_g (eV) of LiH in RS, CsCl, ZB and WZ structures.

	Present work			
Structure	PBE-GGA	mBJ	Experimental works	Other theoretical works
RS	X→X = 3.190	5.200	4.4[34]	4.6723[8], 4.94[33], 5.37[35], 4.92[36]
	W→X = 3.250	5.400		
CsCl	R→X = 0.740	2.940		
	M→M = 4.230	6.480	———————-	1.5[8]
ZB	W→L = 4.340	6.150		
	L→L = 5.240	6.950	————————	————————————————
WZ	Λ→K = 3.800	5.730		
	K→K = 4.250	6.120	————————	————————————————

Table 8. Calculated energy band gap value E_g (eV) of NaH in RS, CsCl, ZB and WZ structures.

	Present work			
Structure	PBE-GGA	mBJ	Experimental works	Other theoretical works
RS	W→L = 3.820	6.700	———————–	4.8560[8], 3.46[37], 5.68[38]
	L→L = 4.630	7.130		
CsCl	R→X = 1.100	4.550		
	X→X = 4.490	7.050	———————-	1.0[8]
ZB	W→L = 3.750	6.500		
	L→L = 4.380	6.900	———————–	————————————————
WZ	Λ→Γ = 3.800	6.600		
	M→M = 4.350	6.800	———————–	————————————————

The calculated PBE-GGA and mBJ-GGA energy band gaps are listed in Tables 7 and 8 along with other experimental [34] and theoretical works [8,33,35–38]. It can be seen that the calculated energy band gaps using mBJ-GGA are broader than those using the PBE-GGA approach. The differences in the energy band gaps between PBE-GGA and mBJ-GGA are 2.01, 2.2, 1.81 and 1.93 eV for LiH in RS, CsCl, ZB and WZ structures, respectively, while for NaH the differences are 2.88, 3.45, 2.75 and 2.80 eV in RS, CsCl, ZB and WZ structures, respectively. Both LiH and NaH in CsCl phase are semiconductors within the PBE-GGA approximation. By using the mBJ-GGA, the LiH and NaH are found wide-band gap semiconductor and insulator, respectively.

It is clear that these compounds, using mBJ-GGA, are insulators in all structures except for LiH in the CsCl structure, where it is a wide-band gap semiconductor. Furthermore, it is evident from Tables 7 and 8 that the energy band gaps for LiH and NaH in the RS structure, using the PBE-GGA approximation, are somewhat smaller than the experimental values. This can be explained as a result of the self-interaction problem contained within the PBE-GGA approximation: due to this unphysical problem, the energy band gap is usually underestimated and sometimes a semiconductor or metallic state is obtained instead of an insulating one. This problem shifts the Li-2s state to an incorrect high energy level and interacts with H-1s state; this interaction increases the coulomb repulsion, which in turn lowers the valence band and causes a narrowing of the energy band gap [39]. Becke and Johnson proposed an exchange potential which was designed to reproduce the exact exchange potential; the calculated energy band gaps by mBJ are in good agreement with the experimental values. Figures 7 and 8 show the density of states (DOSs) for LiH and NaH compounds, respectively, using both the PBE-GGA and mBJ-GGA approaches. In order to understand the nature of these electronic bands' structures further, the total and partial DOSs for LiH and NaH compounds at ambient pressure have also been calculated. DOS of a system describes the number of states at each energy level that are available to be occupied. From the total and partial DOS for LiH (Figure 7) and NaH (Figure 8), one can see that the bands below the Fermi energy level (FE) – indicated by a dotted horizontal line – come mainly from the H-s states along with a small contribution from Li-s, Li-p, Na-s and Na-p states, for the LiH and NaH compounds. Above the FE, the bands mainly come from Li-P and Li-s states for LiH, and from Na-s and Na-p states for NaH, with a small contribution from H-s state for both.

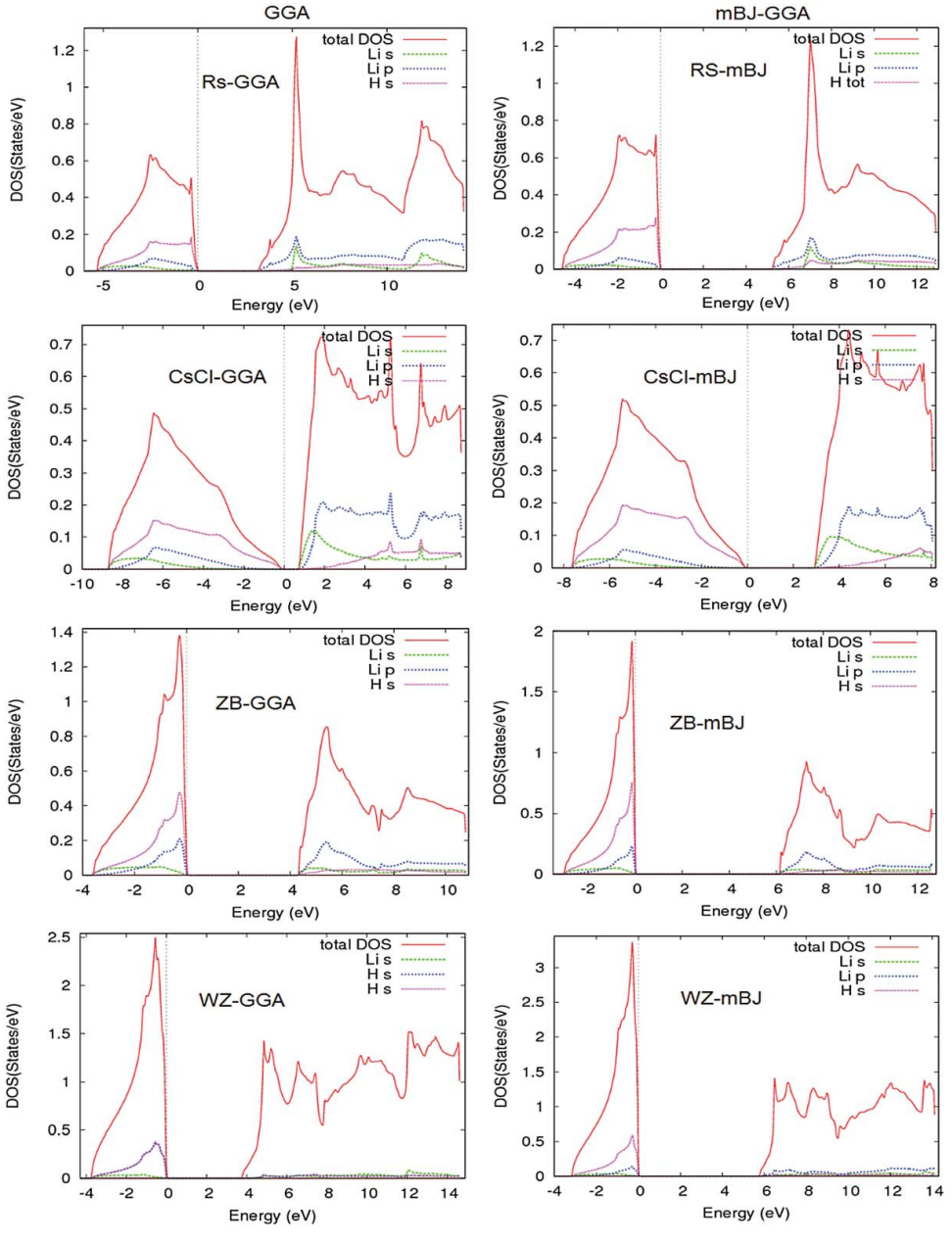


Figure 7. Density of states of LiH in RS, CsCl, ZB and WZ structures.

In the case of the LiH compound using PBE-GGA (mBJ-GGA), the lowest lying bands above FE, at the bottom of the conduction band, are around 3.19 eV (5.2 eV) along the X-point, 0.74 eV (2.94 eV) along the X-point, 4.34 eV (6.15 eV) along the L-point and 3.80 eV (5.73 eV) along the K-point for RS, CsCl, ZB and WZ structures, respectively.

For the NaH compound using PBE-GGA, the lowest lying bands above FE are around 3.82 eV along the L-point, 1.1 eV along the X-point, 3.75 eV along the L-point and 3.8 eV along the Γ-point for RS, CsCl, ZB and WZ structures, respectively. Finally, by using the mBJ-GGA, the lowest lying

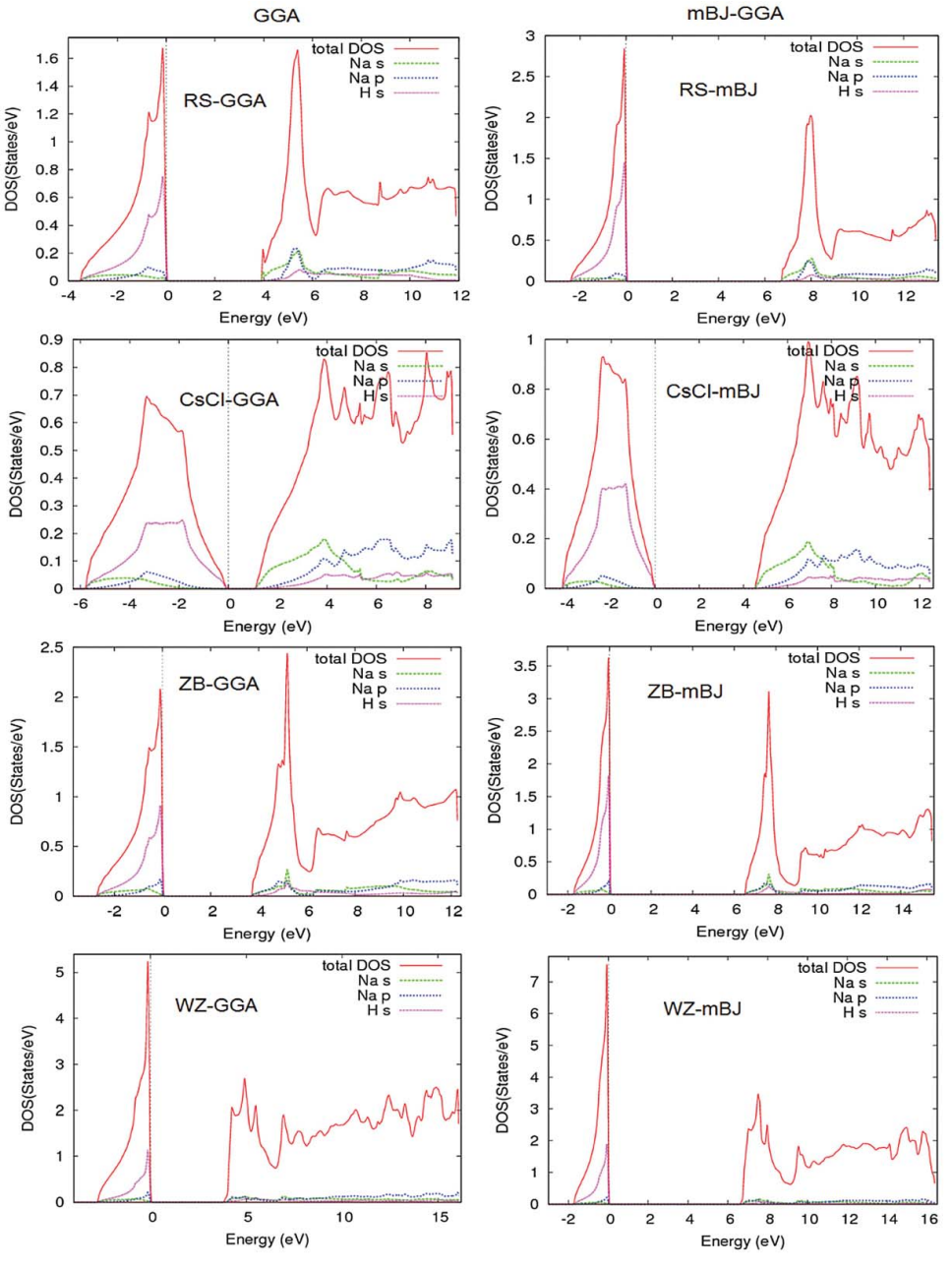


Figure 8. Density of states of NaH in RS, CsCl, ZB and WZ structures.

bands above FE are around 6.7 eV along the L-point, 4.55 eV along the X-point, 6.50 eV along the L-point and 6.6 eV along the Γ-point. Figure 8 also shows that the peaks of the DOSs (states/eV) obtained by using the mBJ approach are sharper and greater in magnitude compared to those obtained by using the PBE-GGA. Figures 7 and 8 that the energy band gap using mBJ-GGA is broader than that using the PBE-GGA method.

4. Conclusions

In this paper, the structural and electronic properties, as well as the structural phase transition of the alkali hydrides XH (X = Li and Na), are calculated in the RS, ZB, CsCl and WZ structures. The FP-LAPW method was used, within the PBE-GGA and mBJ-GGA approaches, as implemented in WIEN2k code. A good agreement with the experimental result was obtained, by applying the mBJ-GGA approach, the energy band gap was modified and thus the CsCl structure was converted from a semiconductor to insulator for LiH and semiconductor to wide energy band gap semiconductor for NaH. The phase transitions predicted in this work are from RS to CsCl, RS to ZB and RS to WZ. It is clear that XH (X = Li, Na), with the PBE-GGA approach, undergoes from RS to CsCl under high pressure of 211.8 and 34.26 GPa for LiH and NaH, respectively. Whereas, under low pressure the RS structure expands and transforms to WZ and ZB. It was also observed that the present calculations agree moderately with other experimental and methods results.

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Disclosure statement

No potential conflict of interest was reported by the authors.

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